

RUN A — PHASE-CHANGE DEMONSTRATION

CIP-BTSB-BT-001 · Supporting Record R-A

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Run date: 29 August 2026 · Final run of the bench-test programme

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

Run A is the **test condition**. It places a 3 mm layer of paraffin wax in the heat path — a material that **does** change phase within the temperature range of the test.

It completes the three-run comparison:

C → B	delay caused by a LAYER BEING IN THE WAY	(thermal resistance)
B → A	delay caused by MELTING	(latent heat)

Runs C and B were performed, recorded and published before this run existed. The pre-registered protocol, predictions, analysis method and falsification criteria were committed with a PGP-verified timestamp on **26 August 2026 at 06:38:48 IST**, before any heated run was performed.

2. CONFIGURATION AS RUN

Item	Detail
Layer	Paraffin wax, 23.00 g, cast solid into the cavity
Cavity	155 × 55 × 3 mm silicone gasket, aluminium foil lined — as Run B
Plates	H 98.24 g, C 97.83 g · pair 172.5 J/K
Heaters	3 × ceramic cartridge in series, 13.7 W electrical
Clamps	4 × G-clamp, 43 / 60 / 105 / 130 mm, finger tight + ½ turn
Insulation	Full ceramic-fibre enclosure, clamps wrapped — as Runs C-2 and B
Orientation	Heated from above

No apparatus change from Run B. The same foil liner, the same clamping, the same insulation build. The wax replaced the oil and nothing else was altered.

3. RESULTS

3.1 Timing and integrity

Power ON	t = 60.0 s · approx. 15:42 clock
Power OFF	t = 2763.7 s · approx. 16:27 clock
Heating duration	45.06 minutes — terminated by the \$5.5 limit
Samples	3,356 · zero sensor faults · 0.00 % fault rate
Peak TC1	89.75 °C at t = 2777.8 s (45.3 min elapsed)
Peak TC2	80.25 °C at t = 2886.1 s (47.1 min elapsed)

Both peaks occur **after** power off, consistent with Runs C-2 and B. TC1 never approached the 110 °C early-stop.

Clock times are reconstructed from timestamped screenshots taken during the run. The elapsed-time values are exact and are the governing figures.

3.2 Baseline — measured before power

TC1 37.025 °C	TC2 37.425 °C	
TC1 - TC2 = -0.400 °C	sd 0.303	n = 60
Ambient (mean plate) = 37.23 °C		

Run B started at **38.11 °C**. The difference is **0.88 °C**, well inside the $\pm 2\text{ °C}$ required by §5.3. **No start-condition deviation is recorded for this run.**

3.3 PRIMARY ENDPOINT — NOT REACHED, REPORTED AS A LOWER BOUND

TC2 peaked at 80.25 °C. **The 85.0 °C endpoint was not reached inside the 45-minute limit.**

t85 (Run A)	>	45.0 minutes
t85 (Run B)	=	37.42 minutes
Δt_latent	>	455 s

The 45-minute limit was not extended. It was set on 26 August in Record R-C, published with its reason, before Runs B and A were performed. Record R-B §8, issued 27 August, stated in advance that this run would probably not reach the endpoint and specified the lower-bound reporting method used here.

The reporting method was fixed before the result existed. It governs.

Falsification test (§9). H1 is falsified if $\Delta t_{\text{latent}} \leq 60\text{ s}$. The measured lower bound of 455 s exceeds that threshold by a factor of **7.6**. **H1 is not falsified.**

4. THE PLATEAU — THE PRINCIPAL QUALITATIVE RESULT

TC1’s climb rate, in two-minute windows, against Run B over the same intervals:

minutes	RUN A	RUN B	
8 – 10	1.12	1.75	
10 – 12	1.50	1.75	
12 – 14	0.38	1.88	
14 – 16	0.87	1.50	
16 – 18	0.37	1.50	
18 – 20	0.88	1.26	
20 – 22	0.50	1.00	
22 – 24	0.62	1.25	
24 – 26	0.75	1.01	
26 – 28	1.12	0.87	← recovered
28 – 30	1.37	0.87	
30 – 32	1.13	0.50	

PLATEAU WINDOW t = 12 to 26 minutes = 14 minutes
TC1 during it 60.50 °C → 69.25 °C

The hot plate’s climb rate fell to **one third** of the control’s while it passed through the paraffin melting range, then **recovered and exceeded the control** once melting was complete. The control shows no corresponding feature at any temperature.

5. ENERGY BALANCE — THE PRINCIPAL QUANTITATIVE RESULT

The energy balance for Run A was integrated using the heat model **fitted to Run B** ($P = 7.26\text{ W}$, $k = 0.1114\text{ W/K}$), and the energy unaccounted for by plate temperature rise was computed:

$$E_{\text{unaccounted}}(t) = \int [P - k \cdot (T_{\text{mean}} - T_{\text{amb}})] dt - C_{\text{plates}} \cdot \Delta T_{\text{mean}}$$

Elapsed	Energy in (J)	Stored in plates (J)	Unaccounted (J)
10 min	3,907	2,458	1,449
18 min	6,383	3,924	2,459
26 min	8,521	4,787	3,735
30 min	9,470	5,606	3,864
34 min	10,287	6,426	3,861
38 min	10,987	7,116	3,871
45 min	11,989	8,064	3,924

MEASURED ABSORPTION	≈	3,900 J
THEORETICAL, 23.00 g	=	4,600 J @ 200 J/g
ENERGY BALANCE CLOSES AT		85 %

The unaccounted energy stops accumulating at approximately minute 30 — the same point at which the rate plateau of §4 ends. Two independent signals, derived by different methods from the same dataset, locate the completion of melting at the

same time.

5.1 Why the sensible heat of the layer does not confound this

The model's P and k were fitted to Run B, which contained **25.6 mL of liquid paraffin oil**. The two layers have near-identical sensible heat capacity:

oil	21.8 g × 2.0 J/g·K	= 43.5 J/K
wax	23.0 g × 2.1 J/g·K	= 48.3 J/K

The sensible-heat term is therefore largely common to both runs and cancels in the comparison. The residual is dominated by latent heat.

5.2 A conservative floor, by a second method

Comparing mean plate temperature directly against Run B, and correcting for the 0.88 °C cooler start:

Peak deficit at 26 min	= 13.87 °C
× 172.5 J/K	= 2,393 J

This method ignores the lower heat loss of a cooler fixture and therefore **understates** the absorption. It is reported as a floor.

MEASURED LATENT ABSORPTION: 2,393 J (floor) to 3,900 J (model)

5.3 Third signature — the pre-specified power method

§6.3 specifies a linear least-squares fit to the mean plate temperature between 40 and 70 °C, converted to power by C_{plates}. Applied without modification:

RUN C-2	7.73 W	
RUN B	5.42 W	
RUN A	2.77 W	← 36 % of the baseline figure

That method assumes all delivered power raises plate temperature. In Run A it does not. **The pre-registered analysis, applied blind, reports the energy as missing.**

6. EARLY-PHASE CONTACT RESISTANCE — A CONFOUND, RECORDED IN FULL

Identified during the run, at approximately four minutes, and reported here in full because it was not anticipated in the pre-registration.

The wax was cast solid into the cavity. Paraffin contracts 10–15 % on solidifying, so the cast layer pulled away from the plate faces and left thin air voids. Air conducts at 0.026 W/m·K. Liquid paraffin oil, by contrast, wets both faces completely.

TC1 – TC2, early phase

minutes	RUN A	RUN B
2	4.00	3.50
4	8.75	6.50
6	12.00	7.50
8	13.00	8.50
10	12.50	7.75
12	12.50	7.50

Run A's temperature drop across the layer is **60 % larger than Run B's before any melting is possible**. Paraffin melts at 58–62 °C; TC1 did not reach 58 °C until approximately minute 10.

6.1 Effect on interpretation

This gap is thermal resistance, not latent heat, and it must not be attributed to the mechanism. It is separable, and it is separated here:

BEFORE 10 min	an elevated TC1–TC2 with NO rate anomaly → contact resistance
12 to 26 min	TC1 rate falls to one third of control, while TC1 passes 60.5 → 69.3 °C → melting

The energy-balance result of \$5 is not affected. Contact resistance changes **where** the temperature sits, not **how much energy the fixture holds**; a resistance stores no energy. The 3,900 J is unaffected by it.

6.2 Corrective action for Gen-0

The PCM must be melted in place, in the assembled fixture, and allowed to re-solidify under clamp load, before

the run begins. One thermal cycle removes the cast voids entirely. This was not done and it cost interpretive clarity in the first ten minutes of the run.

6.3 An unexpected consequence

23.00 g of paraffin at 0.90 g/cm³ occupies 25.6 cm³ against a 25.575 cm³ cavity. On melting, the liquid volume rises to approximately 30 cm³ — 18 % more than the cavity. **No wax was expelled.** The tray and blanket beneath the fixture were inspected after cooling and were clean.

The expansion was absorbed by the cast voids of §6. **The defect and the expansion cancelled.** Both are recorded because neither was predicted.

7. RESULTS AGAINST PREDICTIONS

Quantity	Predicted §7.2	Measured	Verdict
Plateau duration	13 - 25 min	14 min	✓ PASSED
Plateau temperature	58 - 68 °C	60.5 - 69.3 °C	✓ PASSED
t _{as} Run A	17 - 27 min	> 45 min	✗ failed
Δt _{latent}	—	> 455 s	H1 not falsified

7.1 Programme scorecard — all three runs

✗	Run C	delivered power 10–15 W	→ 7.73 W
✗	Run B	Δt _{resistance} 100–250 s	→ 923 s
✗	Run B	TC1–TC2 steady 22–28 °C	→ 7.38 °C
✗	Run B	early stop on TC1 likely	→ did not occur
✗	Run A	t ₈₅ 17–27 min	→ > 45 min
✓	Run A	plateau duration 13–25 min	→ 14 min
✓	Run A	plateau temperature 58–68 °C	→ 60.5–69.3 °C

Five predictions failed and two passed. The two that passed are the two that test the phase-change mechanism. The five that failed are all quantitative predictions about the fixture — delivered power, heat loss, and thermal resistance.

The pre-registered model of the apparatus was substantially wrong. The pre-registered prediction of the mechanism was correct. Both are reported.

8. INTERPRETATION

Three independent lines of evidence, derived by different methods from one dataset, support the same conclusion:

1. A 14-minute suppression of the hot-plate climb rate, inside both the predicted duration and the predicted temperature band, with no corresponding feature in the control.
2. An energy balance in which ~3,900 J entered the fixture and did not appear as plate temperature, ceasing to accumulate at the same moment the rate plateau ends — 85 % of the theoretical heat of fusion of the charge.
3. The pre-specified §6.3 power method, applied blind, reporting 36 % of the baseline delivered power.

Latent-heat thermal buffering across a 3 mm inter-plate layer is demonstrated, quantitatively, in a pre-registered test.

9. LIMITATIONS

Stated in full and without qualification.

1. n = 1. One run per condition. No repeats, no error bars.
2. Substitute material. Paraffin wax, not RT64HC. Different melt point, different latent heat, different chemistry.
3. Latent heat of the wax is a LITERATURE VALUE (200 J/g), not measured. Commercial paraffin ranges 180–220 J/g; against that band the energy balance closes at 77 % to 94 %.
4. One layer of five. Nothing was tested about pyrolytic graphite, expandable graphite, gas capture or dielectric isolation.
5. No cell. Nothing entered thermal runaway. This is a resistive heater and two aluminium plates.
6. The foil bypass (Record R-B §5) carries 44 % of crossing heat and is present in Runs B and A alike.
7. Cast-wax contact resistance (§6) is present in Run A and absent in Run B. It is quantified and separated but not eliminated.
8. The primary endpoint was not reached. t85(A) is a bound, not a measurement.
9. Clock times reconstructed from screenshots, not logged directly.
10. Wax charge of 23.00 g carries a small volume inconsistency (§6.3). At 21.75 g the energy balance would close at 90 % instead of 85 %. The result is robust across the range.

This record demonstrates a thermal mechanism. It says nothing about BTSB as a product, and nothing about battery-cell performance. That is Gen-0.

10. PROGRAMME STATUS

[✓]	GATE 1	software chain	PASSED
[✓]	GATE 2	sensing	PASSED
[✓]	GATE 3	heater circuit	PASSED
[✓]	RUN C	direct contact	t85 = 22.04 min
[✓]	RUN B	liquid paraffin	t85 = 37.42 min
[✓]	RUN A	paraffin wax 23.00 g	plateau 14 min · 85 % closure t85 > 45 min
	Δt _{resistance}	= 923 s	MEASURED
	Δt _{latent}	> 455 s	LOWER BOUND
	H1	NOT FALSIFIED	

CIP-BTSB-BT-001 bench-test programme is complete.

Raw data published as `run_A_20260829.csv`, unedited, alongside `run_C_ABORTED_20260826.csv`, `run_C2_20260826.csv` and `run_B_20260827.csv`.

Record R-A · Issued 29 August 2026 · Final run of the programme. Supports CIP-BTSB-BT-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].